

Little Garden: An augmented reality game for older adults to promote body movement

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ABSTRACT

Physical activity is one of the most effective ways to help older adults stay healthy, but traditional training methods for older adults use single tasks and are boring, often making it difficult for the elderly to achieve good exercise results. In contrast to existing digital games, games based on augmented reality technology have the potential to promote physical activity in the elderly. This paper presents *Little Garden*, an interactive augmented-reality game designed for older adults. It uses projective augmented reality technology, physical card manipulation, virtual social scenarios to increase user engagement and motor initiation. The pilot data show that the game system promotes physical engagement and provides a good user experience. We believe that augmented reality technology provides a new approach to interface design for age-appropriate user-interface experiences.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**.

KEYWORDS

augmented reality, game design, social cognition, older adults

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1 INTRODUCTION

Older adults who engage in regular physical activity are in better health, but they often lack the motivation to stay physically active

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[6]. While the use of traditional digital games has been shown to have positive results in encouraging older adults to increase physical activity, the fixed game flow and single training format may negatively affect their desire to participate.

Recent studies have shown that augmented reality has the potential to help older adults overcome the significant losses and limitations they face in terms of mobility, cognitive abilities, and socialization [3]. For example, Fariba et al. [5] designed a fall prevention training system for older adults using the real-world visibility of augmented reality technology, and a three-dimensional hologram display device proposed by Lee et al. [2] demonstrated that augmented reality can effectively reduce cognitive load and increase interest in training. Augmented reality has demonstrated much potential in age-appropriate design, but the application of a social scenario has been neglected in existing studies, and social engagement is equally important for older adults' health.

In this paper, we propose Little Garden, an augmented reality interactive game for the elderly, which aims to increase the motivation of the elderly to exercise by breaking the boundary between virtual reality and the real world.

2 GAME DESIGN

In this study, we adopted three domains, audio-visual, cognitive, and physical, from Ermi's game design approach[1] to guide the system development. In contrast to traditional digital games, Little Garden has the following three major features.

(1) The use of projected augmented reality. We use the projection mapping method to display the content generated using augmented reality (Figure 1 (c)) on a second information display interface – the projection wall, as shown in (Figure 1 (a)), which provides virtual card content on top of the physical card game, thus effectively expanding the user's game space.

(2) Tangible toolkit of cards. The user participates in a virtual social scenario by controlling a physical card in the game (Figure 1 (b)), and interacting with the recognition camera in front of them. After entering the social scenario, the system automatically plays an animation, and the user can control the position of virtual objects and characters using the card in their hand. Each social scenario animation last for 2 minutes, and the total game length is about 15 minutes.



Figure 1: Little Garden: (a) game system setup, (b) tangible toolkit, (c) social scenario display, and (d), (e) users performance

(3) Co-design with the elderly. 1) Social scenarios: To design the game story, we referred to the hobbies of 67 elderly people (29 male and 38 female) gathered using a preliminary questionnaire (Table 1), and we selected the higher-ranked hobbies to create the background of the story design. 2) Voice commands: The game is controlled using voice commands corresponding to the social scenarios. These voice commands were recorded by the elderly co-designers and integrated into the voice module of the Unity engine.

3 TECHNICAL SYSTEM

The system consists of a deck of cards, projection screen, camera, and demonstration application implemented using Unity, as shown in Figure 2. When the game is running, the Vuforia augmented reality engine recognizes the content of the cards, and the game characters and corresponding social scenarios appear on the interface. When the user holds up the card for content recognition, it transfers the information to the database for content comparison and recognition.

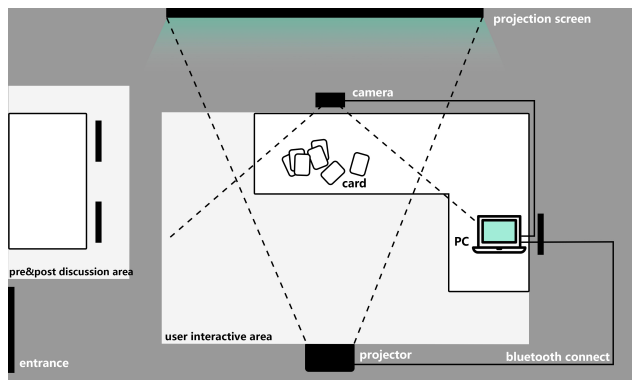


Figure 2: Structure of the game

4 EVALUATION

We recruited 12 participants (mean age = 69 years, standard deviation = 2.65, 5 males and 7 females) to play the game in this study. The Qingdao University Ethics Review Committee reviewed and approved the study involving human participants, and the subjects themselves signed a written informed consent form for the game.

We collected data from three main sources: 1) pre-experimental MoCA (Montreal Cognitive Assessment Test), 2) experiment video recordings, and 3) post-test questionnaires. We used the cognitive test data to analyze the group of participants, and the other two sources of data to analyze the usability and user experience of the game system. The correlation analysis showed that there was no significant correlation between the MoCA scores and the total scores with respect to the age or gender of the participants, indicating that there was little individual difference in the experimental group of older adults.

All participants completed the full game. We observed that participants T1, T4, and T12 swayed their bodies and hummed along with the music throughout the experiment after the dance card was recognized. Participants T9 and T12 both expressed that this game was interesting and wanted to experience it again, and T12 asked the staff during the break whether she could bring her grandchild to play with her next time. T11 found the cards with moving ability more appealing (Figures 1 (d) and (e)). Some participants made negative comments about this game; for example, criticizing the system delay in the card recognition process (a common problem in the field of computer vision [4]). Overall, the results show that Little Garden can promote physical activity and increase user motivation, giving us a glimpse of the potential of augmented reality technology in game design for older adults.

5 CONCLUSION

Our preliminary results show that Little Garden has the potential to increase the physical engagement and motivation of older users. In future work, we will conduct long-term tracking tests and increase

Table 1: Pre-test questionnaire

Rank	Hobby	Ratio
1	walk	66.67%
2	work out	40.74%
3	sing and dance	37.04%
4	play poker,mahjong	37.04%
5	grow flowers	22.22%
6	travel	22.22%
7	fishing	11.11%

the number of users to collect more accurate data to assist us in the iterative development of the system. For example, we will allow participants to customize virtual characters and social stories as well as develop multiplayer modes.

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REFERENCES

- [1] Adrian David Cheok, Kok Hwee Goh, W. Liu, Farzam Farbiz, Sze Lee Teo, Hui Siang Teo, Shang Ping Lee, Yu Li, Siew Wan Fong, and Xubo Yang. 2004. Human Pacman: a mobile wide-area entertainment system based on physical, social, and ubiquitous computing. In *ACE '04*.
- [2] I-Jui Lee, Chien-Hsu Chen, and Kuo-Ping Chang. 2016. Augmented reality technology combined with three-dimensional holography to train the mental rotation ability of older adults. *Comput. Hum. Behav.* 65 (2016), 488–500.
- [3] Li Na Lee, Mi Jeong Kim, and Won Ju Hwang. 2019. Potential of Augmented Reality and Virtual Reality Technologies to Promote Wellbeing in Older Adults. *Applied Sciences* (2019).
- [4] Jiajia Li, Zixia Zheng, Xiemin Wei, and Guanyun Wang. 2021. FaceMe: An Augmented Reality Social Agent Game for Facilitating Children's Learning about Emotional Expressions. *The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology* (2021).
- [5] Fariba Mostajeran, Frank Steinicke, Oscar Ariza, Dimitrios A. Gatsios, and Dimitrios Ioannis Fotiadis. 2020. Augmented Reality for Older Adults: Exploring Acceptability of Virtual Coaches for Home-based Balance Training in an Aging Population. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (2020).
- [6] Frank J. Penedo and Jason Dahn. 2005. Exercise and well-being: a review of mental and physical health benefits associated with physical activity. *Current Opinion in Psychiatry* 18 (2005), 189–193.